

Bilaga A

VisCon - Manual

A.1 Background

A.1.1 Aim with program

VisCon was developed in order to facilitate the understanding of two dimensional consolidation. The program visualises how the excess pore water pressure varies with time. It also allows the user to extract one dimensional plots of both excess pore water pressure and effective stress at any defined time step. These plots can be extracted vertically or horizontally at any defined section of the model. The program is intended to be used as a complement to ordinary course literature in order to enhance the understanding of consolidation in two dimensions.

A.1.2 Consolidation

When a ground surface is subjected to a load an increased level of stress will occur in the soil underneath. As soil material has a limited permeability, an increased stress will initially result in an excess pore water pressure. However, as time passes the water will flow and the pore water pressure will even out. The relation between the total stress $\Delta\sigma_z$, the excess porewater pressure u_e and the effective stress $\Delta\sigma'_z$ is given by the effective stress equation:

$$\Delta\sigma_z = u_e + \Delta\sigma'_z \quad (\text{A.1})$$

The total change in stress will always equal the sum of the excess pore water pressure and the effective stress. When the pore water pressure reaches its static condition,

i.e. $u_e = 0$ the soil is considered consolidated and the total change in stress equals the effective stress, $\Delta\sigma_z = \Delta\sigma'_z$.

For further theoretical background the user is directed to the master's dissertation in which this program was developed. Non-Swedish-speaking users are directed to the following literature:

- Terzaghi, Karl: *Theoretical soil mechanics*. John Wiley & Sons, New York, 1943.
- Craig, Robert F: *Craig's soil mechanics*. CRC Press, London, 2004
- Tudisco, Erika och Dahlblom, Ola: *Course literature in Foundation Engineering*. LTH, Lund University, Lund, 2017.

A.1.3 Problem statement

Assumptions and limitations

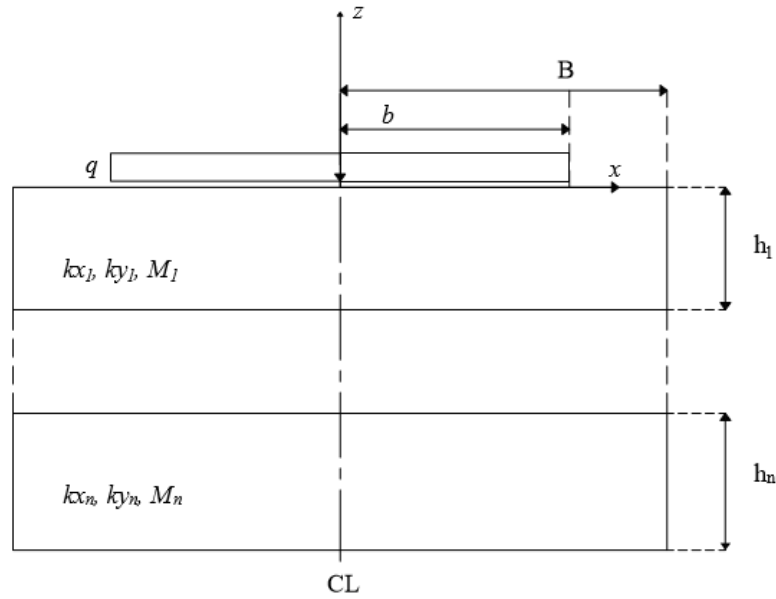
This program facilitates a numerical solution of the two dimensional consolidation equation. As the program is limited to solve two dimensional problems, it can only be applied when the groundwater flow is limited to the xz -plane. In other words, the program solves problems with a load of which the length is greater than the width. Within civil engineering such an application could typically be modelling a soil profile underneath a road embankment. However, it could also give an accurate approximation of the consolidation underneath a long building. This version of the program is limited to a symmetrical case with uniformly distributed load.

In solving the differential equation of consolidation, a number of assumptions are needed. The ones used in this program are listed below:

- The soil is fully saturated.
- Darcy's law is applicable.
- The pore water, as well as the solid grains, are considered incompressible at a material level.
- The process of consolidation is dependent only on the delay of pore water flow described by Darcy's law.

Model

The model used in the program is shown in Figure A.1. The program allows the user to analyse a soil profile with an unlimited number of soil layers. The user can adjust the depth and soil characteristics within each layer. Furthermore, the user is allowed to adjust the magnitude and width of the load, as well as the width of the soil examined.



Figur A.1: *The model for which the program solves the differential equation of consolidation.*

A.1.4 Implementation

The visualisation tool is developed in Python with the use of VisVis, PyQt, Numpy and CALFEM for Python. It uses the finite element method in solving the differential equation describing the process of consolidation. It is a numerical solution involving a time approximation according to the backward difference method.

A.2 How to use VisCon

A.2.1 Tab 1: Model

Overview

When the program is initiated the user interface shown in Figure A.2 will appear. The controls are updated with a pre-defined model.

The screenshot shows the 'Model' tab of the VisCon software. The interface includes the following sections:

- Model:** A diagram showing a cross-section of a soil layer with a central load q applied over a width b . The total width of the model is B . The soil is divided into two layers with properties kx_1, ky_1, M_1 and h_1 for the top layer, and kx_2, ky_2, M_2 and h_2 for the bottom layer. The center line is labeled 'CL'.
- Soil parameters:** A section for defining soil properties. It includes a text input for 'Pore water unit weight [N/m^3]' with a value of 10000.0. Below this are 'Add layer' and 'Remove layer' buttons. A table lists the parameters for the two layers:

	h [m]	kx [m/s]	ky [m/s]	M [Pa]
1	15.0	1e-09	1e-09	900000.0
- Geometry and load case:** A section for defining the geometry and load. It includes text inputs for 'B [m]' (8.0), 'b [m]' (2.0), and 'q [Pa]' (30000.0).
- Boundary conditions:** A section with a checkbox labeled 'Permeable ground' which is checked.
- Element size:** A section with a slider control for 'Element size' set to 0.5.
- Preview of geometry, mesh and initial stress distribution:** A large area at the bottom containing three gray rectangular placeholders for the model preview, mesh, and stress distribution. An 'Update preview' button is located at the bottom center of this section.

Figur A.2: The first interface of the Main Window, intended for defining the model.

Soil parameters

In section Soil parameters, shown in Figure A.3, the user is allowed to enter desired soil parameters. The first field is intended for the pore water unit weight, γ_w , given by $\gamma_w = \rho_w g$ [N/m³]. The pore water density, ρ_w , varies with the level of dissolved salts in the water, but is normally given by $\rho_w = 1000$ kg/m³. The gravitational acceleration, g , is given by $g = 9.81$ m/s², i.e. the unit weight is $\gamma_w = 1000 \cdot 9.81 = 9810$ N/m³, or slightly higher if there is dissolved salts. Often $\gamma_w = 10\,000$ N/m³ can be assumed.

Soil parameters:

Pore water unit weight [N/m³]

Add layer
Remove layer

	h [m]	kx [m/s]	ky [m/s]	M [Pa]
1	15.0	1e-09	1e-09	900000.0

<

>

Figure A.3: Section to assign parameters for pore water and soil profile.

Next field in Soil parameters is intended for the soil profile. The user may adjust the number of soil layers by adding or removing layers to the table. For each layer it is possible to adjust depth h [m], permeability in x -direction k_x [m/s], permeability in z -direction k_z , and compression modulus M [Pa].

Geometry and load case

In section Geometry and load case, shown in figure A.4, it is possible to determine the geometry and load case. The model is symmetrical and the dimensions given are with reference to the centre line shown in Figure A.1.

B [m] is the width of the soil examined. It is important to choose a width of the soil considerably larger than the width of the load. The reason behind this is that the boundary condition implemented in this program assumes the flow perpendicular to the line opposite to the centre line to be zero. This boundary could also be represented by an impermeable structure surrounding the soil profile. b [m] is the width of the load in the z -direction. The load is assumed to be uniformly distributed

Geometry and load case:

B [m]

b [m]

q [Pa]

Figure A.4: Section to assign geometry and load case.

and infinite in the y -direction. The magnitude of the load is adjusted by the value of q [Pa].

Boundary conditions

The boundary conditions determine along which surfaces the pore water pressure should be zero at $t > 0$. The boundary conditions for the top surface are pre-defined and assumed to be zero, as water is free to flow perpendicular to its direction. It is not possible to change this condition in the program.

Boundary conditions:

☒ Permeable ground

Figure A.5: Section to assign boundary conditions.

In section Boundary conditions, shown in Figure A.5, it is possible to choose if the bottom surface should be permeable or not. If the check box is ticked the bottom surface is permeable, allowing water to flow perpendicular to its direction.

Element size

In section Element size it is possible to adjust the element size by controlling the slider, shown in Figure A.6. It is important to have a fine mesh since it results in a more precise solution. However, a fine mesh also results in large matrices and subsequently more time consuming calculations.

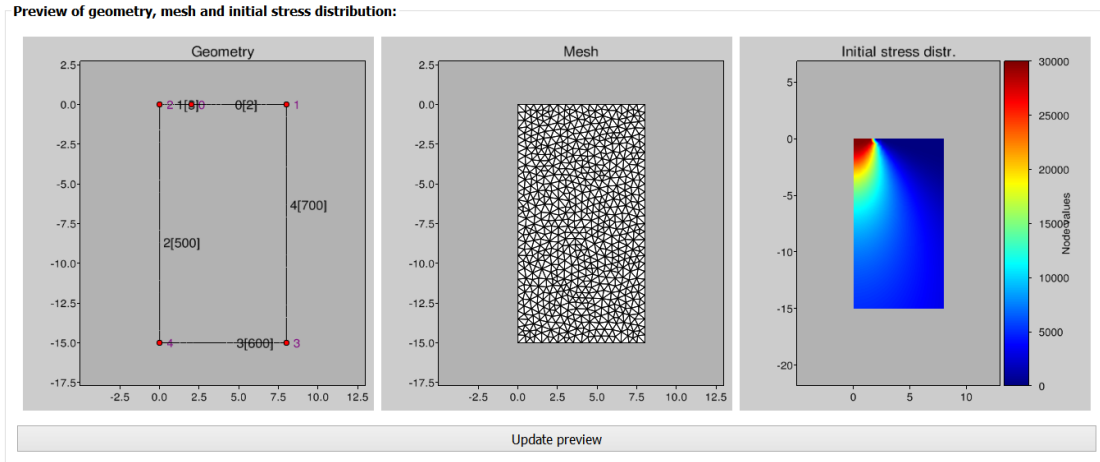
Element size:

0.5

Figure A.6: Section to control element size.

Preview

In section Preview, shown in Figure A.7, it is possible to preview geometry, mesh and initial stress distribution based on given input data. In order to update the preview, if any values have been changed, press the button *Update preview*.

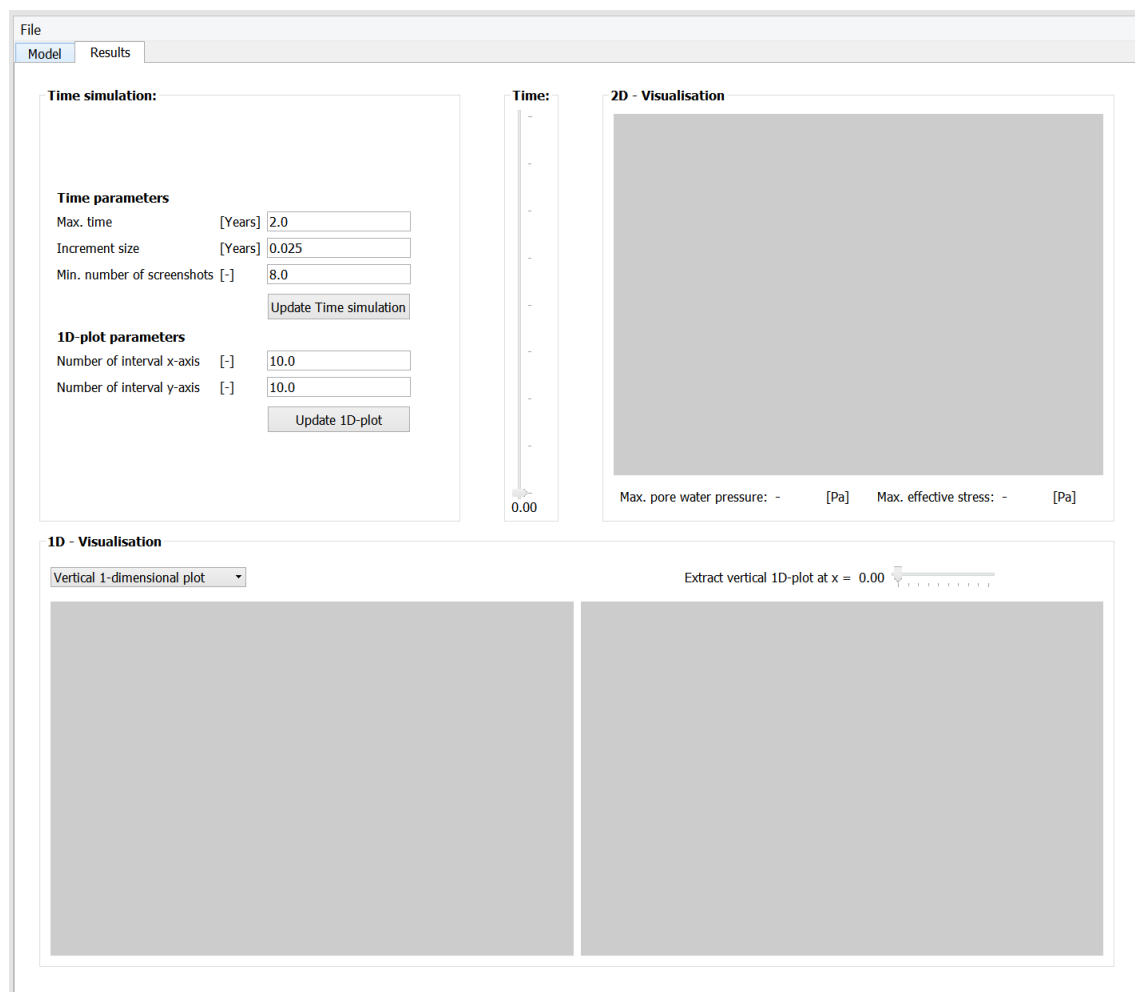


Figur A.7: *Preview based on given values for the model.*

A.2.2 Tab 2: Results

Overview

When the user clicks on the second tab, Results, the program will enter calculation mode. It will perform the meshing, generate the finite element matrices and perform a time simulation based on pre-defined values. Following, the user interface shown in Figure A.8 will appear.



Figur A.8: *The second interface of the Main Window, intended for defining parameters for the time simulation as well as visualising the results.*

Time simulation

It is possible to control the time simulation by varying the values in section Time simulation, shown in Figure A.9. In the first part of this section the time parameters

are given. It is possible to decide how long the time simulation will be as well as the increment size of each time step. It is important not to choose a too large increment since it may result in a too rough time approximation. However, a small timestep will result in more time consuming calculations.

Furthermore, it is possible to choose the minimum number of screenshots for the time simulation. The program will store and plot values of the excess pore water pressure for this number of time steps. The time steps will be equally distributed over the simulated time .

Time simulation:

Time parameters

Max. time [Years]

Increment size [Years]

Min. number of screenshots [-]

Update Time simulation

1D-plot parameters

Number of interval x-axis [-]

Number of interval y-axis [-]

Update 1D-plot

Figur A.9: Section to enter values for the time simulation as well as the visualisation.

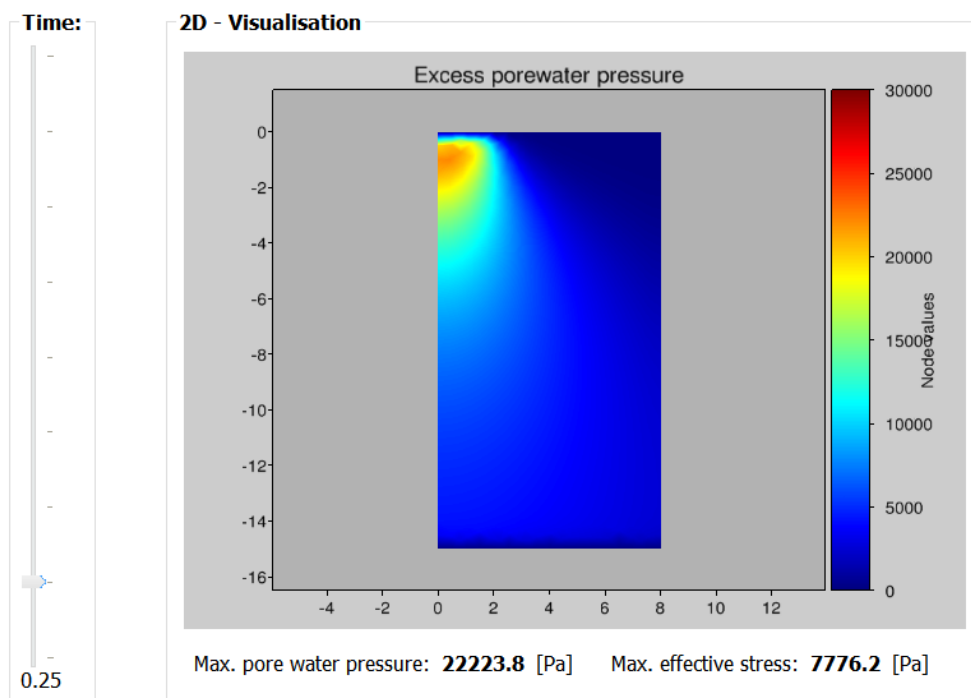
In the second part of this section it is possible to choose how many x - and z -values the program should extract one dimensional plots for. Based on the number the program will generate as many one dimensional plots, equally distributed over the x - and z -axis. The number connected to x will generate vertical one dimensional plots and the number for z will generate horizontal one dimensional plots.

Time

With this control it is possible to vary for which time step the current two and one dimensional visualisations are shown.

2D - Visualisation

In this section, shown in Figure A.10, the results of the transient analysis of the excess pore water pressure will be shown in two dimensions. The two dimensional figure shows the excess pore water pressure, plotted with a colour bar visible to the right side. Next to this figure a table gives the maximum excess pore water pressure and the maximum effective stress. According to the effective stress equation the sum of these will equal the total stress at all time steps.



Figur A.10: A two dimensional visualisation of the excess pore water pressure at a defined time step.

The two dimensional colour plot and the maximum values will change accordingly to the current time step, defined in section Time.

1D - Visualisation

In the section for one dimensional visualisation, shown in Figure A.11 two one dimensional plots are extracted from the calculated excess pore water distribution. It is possible to decide if the one dimensional plot should be a vertical or horizontal section in the drop-down menu. By the slider it is possible to choose for which x - or z -value the section should be plotted.

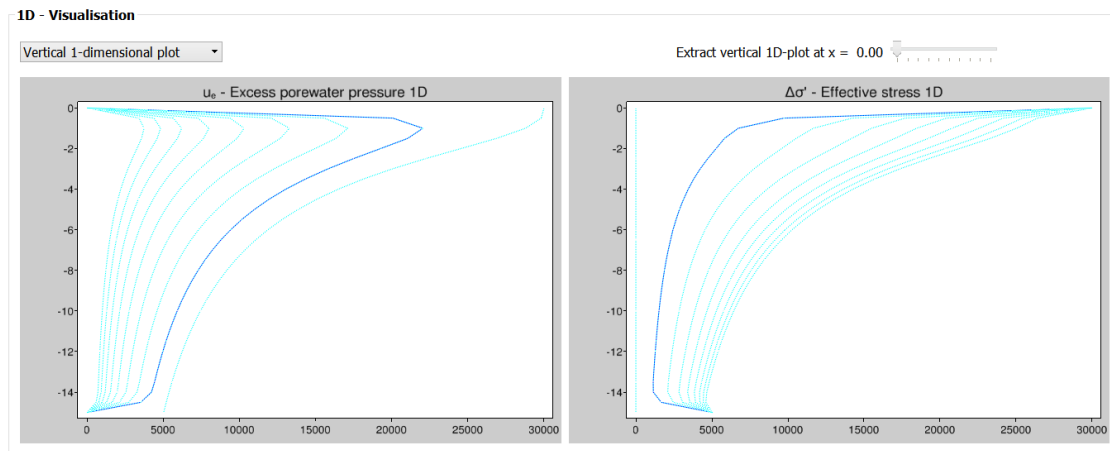


Figure A.11: *Extracted one dimensional plots, showing excess pore water pressure and effective stress at a defined section of the ground model.*

The first figure shows the excess pore water pressure at a chosen section, and the second figure shows the effective stress. A number of graphs will be visible, representing all defined plot time steps. The current time step is highlighted and it changes accordingly to the time step defined in section Time.

A.2.3 File manager

New

Open a new file, with pre-defined values of the ground model as well as the time simulation.

Save

Save the current model.

Save as

Save the current model, giving it a new file name.

Open

Import an already existing model.

Exit

Exit the program.